Assessment of beat-to-beat heart rate detection method using a camera as contactless sensor

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***Abstract*— Video photoplethysmography (videoPPG) has emerged as area of great interest thanks to the possibility of remotely assessment of cardiovascular parameters, as heart rate (HR), respiration rate (RR) and heart rate variability (HRV). The present article proposes a fully automated method based on chrominance model, that selects for each subject the best region of interest (ROI) to detect and evaluate the accuracy of beat detection and interbeat intervals (IBI) measurements. The experimental recordings were conducted on 26 subjects which underwent a rest-to-stand maneuver. The results show that the accuracy of beat detection is slightly better during supine position (95%) compared to the standing one (92%), due to the maintenance of the balance that introduces larger motion artifact in the signal dynamic. The error in the measurement (expressed as *mean*$\pm$*sd*) of instantaneous heart rate is of +0.04 $\pm$3.29 bpm in rest and +0.01$\pm$4.26 bpm in stand.**

I. Introduction

Recently a new method has been developed to assess the blood volume pulse signal (BVP) remotely by camera recording of human body [1]–[3]. The physical principle is based on the study of reflected light from superficial arteries that is carrying on the pulse wave information.

The possibility to record remotely videoPPG signal makes this technology useful in telemonitoring, fitness and burn injuries where the use of leads is critical. Moreover combined with facial features analysis, it could let to study emotional states of subjects during cognitive stress.

The main disadvantages explored in literature are linked to motion artifact that reduces the signal-to-noise ratio (SNR) of the desired signal. Many methods were proposed: Poh *et al.* [2] applied the ICA to the sources signals in order to capture and isolate the PPG dynamic, while Lewandoska *et al.* [4] focused on PCA method demonstrating its computational speed compared to ICA method; the work of De Haan *et al.* [5] on the contrary, proposed a chrominance method based on the study of scattered light.

The physiological parameters which can be extracted from videoPPG, include the heart rate (HR), respiration rate and the oxygen saturation [6], [7]. However, to the best of our knowledge, few researchers [8]–[10] have focused on the study of interbeat intervals (IBI) series that could provide a more detailed hemodynamics analysis.

The objective of this paper is twofold: 1) using the chrominace model proposed by De Haan *et al.*, with the development of an algorithm, which analyzes data from different ROI and automatically selects the best ROI for the extraction of videoPPG signal; 2) to quantify the accuracy of beat-to-beat analysis applied on videoPPG signal compared to the gold standard of ECG during two conditions: *supine* and *stand* position.

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II. Methods

A. Experimental setup

Twenty-six volunteers were recruited for rest-to-stand experiment (12 females and 14 males) with different ages (20-36) and diverse skin color. They received a comprehensive written informed consent to sign. The experiment was conducted in accordance with the Helsinki Declaration.

The camera used to record the video was a GigE Sony XCG-C30C. The video data output was exported in RGB uncompressed raw format with a frame rate of 60 Hz. The camera was equipped with 15 mm fixed focal length lenses (Tamron 25 HB), to increase the region occupied by the partecipant's face.

Three leads ECG signal (FlexComp Infiniti by Thought Technologies, Inc.) was acquired. The electrodes were positioned on the body surface of the subject according to the manufacturer's manual: the negative electrode was placed on the right shoulder, the positive electrode on left side of the chest and the ground electrode on the left shoulder.

The experiment was conducted indoor, with a changing mixture of sunlight passing through the window and artificial light provided by illumination.

The experimental setup included a bed, positioned perpendicular to the window in order to have direct sunlight in front of the subject. A tripod holding the camera, was positioned alongside the bed. The tripod was equipped with a mechanical arm that allowed the manual movements of the camera during the rest-to-stand maneuver.

During the whole session the ECG and video recordings were acquired, and they were synchronized by an external software. The experiment included two sessions:

Rest session: the subject was asked to lay on the bed facing the camera. He was instructed not to move the head in order to reduce movement artifacts. After a time period of 2-3 minutes needed to relax the subject, the synchronized measurements of video recording and ECG signal were started for a period of 5 minutes;

Stand session: the subject was asked to get up and to stay in front of the camera facing it. During the first 30 seconds, the ECG measurements and video recordings were suspended, to let the operator move the camera and the subject to take the

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standing position. After this period, the ECG and the video recording started for a period of 5 minutes.

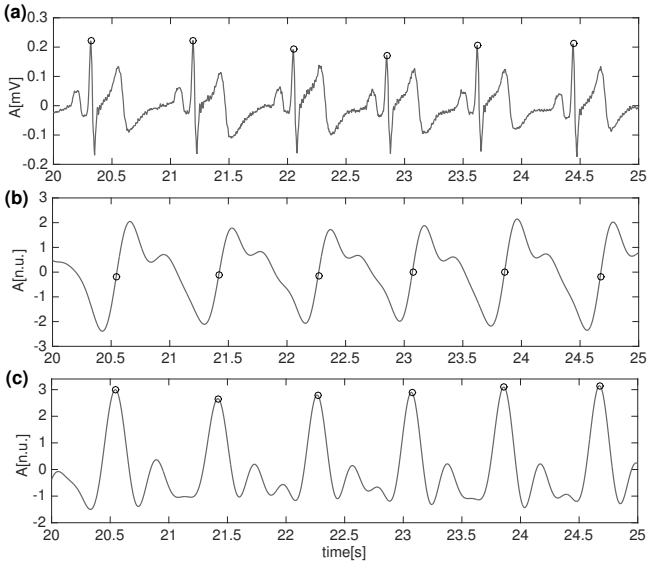


Fig. 1: Example of ECG and BVP signals measured by three leads ECG sensor and digital camera for a time window of 5 sec. (a) ECG waveform with R peaks detection; (b) BVP waveform with estimation of beat detection as maximum derivative of the anachrotic phase; (c) first order derivative of BVP waveform with detected peaks locations.

B. Extraction of videoPPG signal

The video recordings were saved in RGB raw format and processed offline to derive the videoPPG signal. Three regions of interests (ROI) were considered: *forehead* (ROI_1), *nose* (ROI_2) and *cheek* (ROI_3). The ROI detection was developed in OpenCV by the combination of the Viola-Jones face detection algorithm and the Lucas-Kanade-Tomasi motion flow tracking algorithm (LKT).

A spatial average of pixels intensity was calculated for each ROI in order to have $N_{channels} \times N_{ROI}$ time series. The temporal series were first detrended using the tecnique described on [11] setting the smoothness parameter $\lambda = 400$. A hamming bandpass filter with cutoff frequencies of $f_1 = 0.1$, $f_2 = 8Hz$ was applied to remove high frequencies correlated mainly to artificial light and low frequencies associated to slow movements.

For the extraction of the cardiac pulse wave, the method proposed by deHaan *et al.* was applied. It can be resumed in three aspects:

a) Conversion of RGB space color to chrominance model to eliminate specular reflection component due to the light source and to enhance blood volume changes;
b) Correction for inter-subject skin-tone variability by assuming a standard skin tone for everyone;
c) The videoPPG signal is extremely sensible to motion artifact. Therefore, assuming that the movement affects the channels equally, the ratio between the chrominance vectors is considered as robust estimation of videoPPG.

Considering these assumptions, for each ROI it is possible to maximize the pulsatile component of the reflected light and minimize extraneous noise by an appropriate weighing of R, G and B components:

$$S = c_1 * R + c_2 * G + c_3 * B \tag{1}$$

where $c_1 = 3 * \left(1 - \frac{\alpha}{2}\right)$, $c_2 = -2 * \left(1 + \frac{\alpha}{2}\right)$, $c_3 = \frac{3*\alpha}{2}$ are the linear coefficients and $\alpha = \frac{\sigma(X)}{\sigma(Y)}$ is the normalization factor that takes into account different amplitudes of the two chrominance signals X and Y. Therefore three videoPPG signals S_1, S_2 and S_3 were obtained for each ROI. Between the N_{ROI} temporal series, the selection of the best videoPPG signal was achieved by calculating the SNR of the power spectral density (PSD) of the sources by the following formula:

$$SNR = 10 * \log_{10}\left(\frac{\int_{f_1}^{f_2} \mathrm{PSD}(f)\,\mathrm{d}f}{\int_{0.1}^{f_1} \mathrm{PSD}(f)\,\mathrm{d}f + \int_{f_2}^{4} \mathrm{PSD}(f)\,\mathrm{d}f}\right) \tag{2}$$

where $f_1 = f_p - 0.15$, $f_2 = f_p + 0.15$ and f_p is the pulse frequency (measured in Hz) identified as the highest peak in the PSD. Between the three videoPPG signals associated to each ROI, the algorithm chooses the one that presented the highest SNR.

During the standing period, a subset of subjects moved their head, compromising the quality of the signal. To be more robust to motion artifact, the normalization of the coefficients that weight the R, G and B components of each ROI was conducted over the windowed signal. As in the article De Haan *et al.* the Hann window was applied for individual intervals with a 50% of overlapping. The time interval of each window was considered of 3 seconds.

C. ECG and videoPPG peaks detection

To improve the peak detection procedure, the videoPPG signal was smoothed by a filterbank with fundamental frequency centered around the pulse frequency and considering one harmonic of the cardiac pulse. Then videoPPG was interpolated using cubic splines and resampled at 256 Hz, to match the temporal resolution of the ECG signal.

The peaks detection on videoPPG signal was done on the first derivative of the signal applying the algorithm proposed by Scholkmann *et al.* [12]. In this way the position of each peak corresponds to the maximum derivative on the rising edge of the cardiac pulse (defined as *anachrotic phase* [13]). To extract the ECG peaks, the method proposed by Alfonso *et al.* [14] was applied, and a visual inspection of the detection correctness was done to ensure a gold standard for the analysis. Fig. 1 shows an example of ECG, videoPPG and its first derivative of a subject during rest condition for a temporal window of 5 seconds. Fig. 1(a) shows the ECG signal with detected R peaks, while Fig. 1(b) the videoPPG signal with positions corresponding to the peaks of the first derivative (showed in Fig.1(c)).

The main temporal difference between ECG and videoPPG beats position is the time that the pulse wave needs to travel from the heart to the face, called pulse transit time (PTT). This time is measured as the difference between the R-wave on the ECG and the peak value detected on the videoPPG. In

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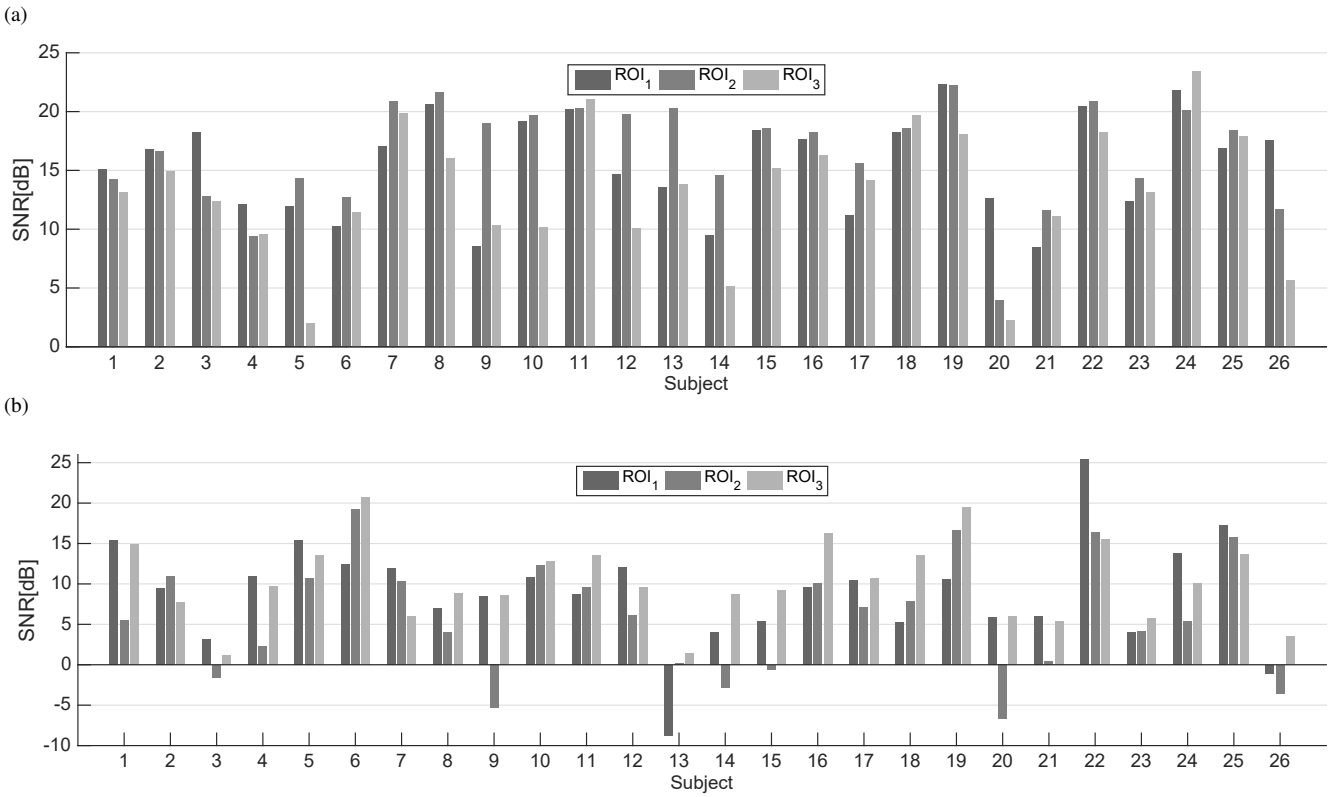


Fig. 2: Comparison of SNR obtained for all subjects and different facial regions (ROI_1,ROI_2, ROI_3). (a) Rest session; (b) Stand session.

order to conduct the beat-to-beat comparison between ECG and videoPPG signals, a temporal alignment between the two series of peaks positions was applied. Assuming constant the PTT during the five minutes of recordings, mean PTT was subtracted to videoPPG beats positions.

D. Quality assessment

The performance of beat detection method was measured by means of *sensitivity* (Se), *positive predictivity* (+P) and *accuracy* (Acc) parameters. Each videoPPG beat position was considered as true positive (TP) if it occurred within 150 ms to an annotated beat in the reference ECG. Detections that occurred outside the temporal range or double detections were considered as false positive (FP); while missing detections were considered as false negative (FN).
The Bland-Altman analysis was conducted in order to evaluate the agreement of instantaneous pulse rates (calculated every three cardiac cycles) between the videoPPG method and the reference signal of ECG during rest and stand conditions.

III. Results

Tab. I shows the averaged results of Se, +P, and Acc of heart beat detection measured in rest and stand conditions. Although the results show a decrease in all parameters during the standing position, there is no significant difference between the two conditions in terms of Se, +P and Acc. Our results are comparable to the ones obtained by Zaunseder *et al.* [10]. However their evaluation was conducted only on 60 s of video recording during rest session, while our analysis is referred to 5 minutes during two different sessions. Figs. 2(a),(b) show the SNR obtained for each subject in different ROI during rest and stand conditions respectively(SNR_{rest}, SNR_{stand}). There is a clear degradation of SNR_{stand} compared to SNR_{rest}, as confirmed in Tab. II where the averaged decrease of SNR is reported for each ROI during rest-to-stand transition. The explanation is linked to the presence of greater artifact movement during the stand period. In fact, the subject, in the attempt to maintain his balance, is moving his head greater than the rest position where the head is more stable. During the rest condition, the ROIs with highest SNR are the ROIs referred to the forehead and nose, while in stand position the ROI associated to the cheek was more robust compared to the other ROIs. However, in our opinion (confirmed by Fig. 2), it is not possible to consider a particular ROI as reference for the extraction of videoPPG signal, since the optimal selection appears to be inter-subject and posture dependent.

TABLE I: Mean beat detection accuracy according to *Se*, *+P* and *Acc* over all subjects for rest and stand conditions.

Condition	Sensitivity	Positive predictivity	Accuracy
	(all values are expressed in *mean*± *std*)		
Rest	0.995 ± 0.007	0.989 ± 0.016	0.984 ± 0.021
Stand	0.992 ± 0.008	0.986 ± 0.017	0.979 ± 0.023

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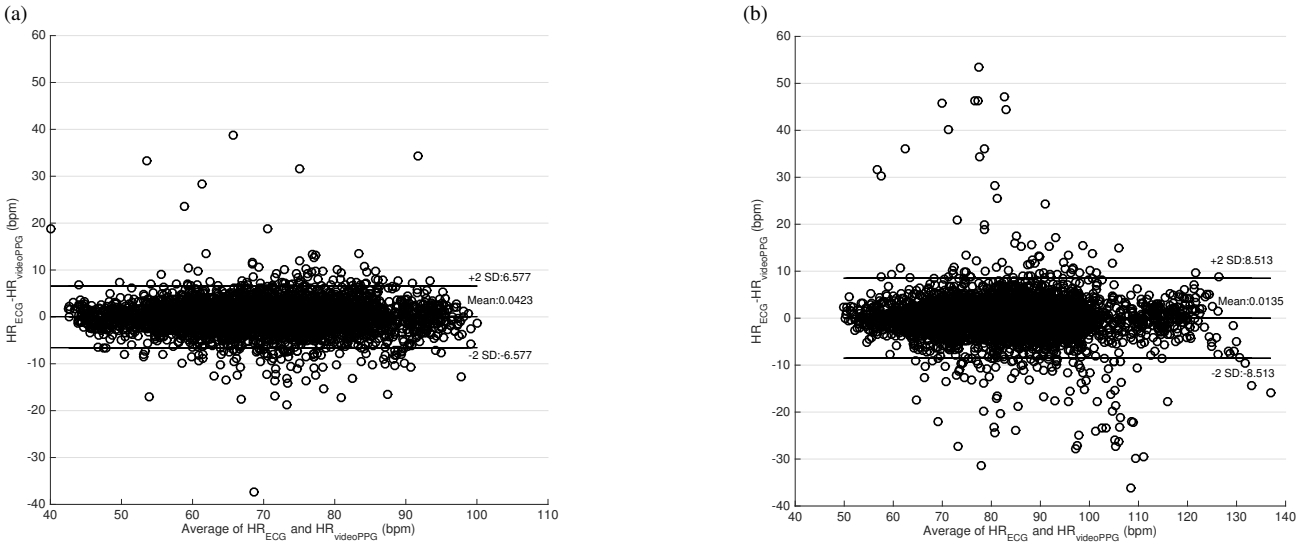


Fig. 3: Bland-Altman plot illustrating the agreement between instantaneous heart rate (calculated every three cardiac cycles) measured by the reference ECG sensor and digital camera. (a) Rest session; (b) Stand session.

TABLE II: Variation of mean SNR for each ROI from rest to stand session.

	$ROI_1(dB)$	$ROI_2(dB)$	$ROI_3(dB)$
ΔSNR	-6.63	-10.61	-3.02

Bland-Altman plots in Figs. 3 show the accuracy of instantaneous pulse rate measurements of the whole population comparing the videoPPG method with the reference ECG respectively during rest (3(a)) and stand (3(b)) position. In each Bland-Altman plot the centered line represents the bias and the other two lines represent the 95% limits of agreement. During rest period the bias is equal to $+0.042$ bpm, with 95% of agreement inside ± 6.577 bpm. During the standing condition the bias is $+0.013$ bpm, while the 95% of instantaneous pulse rate values are contained in the range ± 8.513 bpm.

Considering the data presented in Tab. I and in Figs. 3, it is possible to assert that, despite the decrease of the quality of the videoPPG signal during the standing position, the heart beat detection accuracy is not significantly affected by subject position; moreover, the measure of mean and standard deviation of instantaneous pulse rate is unchanged between rest-to-stand maneuver.

IV. Conclusion

We have presented a fully automatic method based on chrominance model to extract single beats using videoPPG signal. Despite the SNR decreasing, the results have shown that the beat-to-beat accuracy and instantaneous heart rate measurements do not change significantly by modifying subject's posture. Future work should test the robustness of the presented method with the enlargement of subjects number and setting a more realistic environment without controlled head movements.

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